

AMTAS-5/10

Amlodipine Besylate Tablets USP 5 mg and 10 mg

FOR SPECIALIST'S USE ONLY

Active Ingredient:

Amlodipine Besylate

The uncoated tablets contain amlodipine besylate USP equivalent to 5 mg and 10 mg amlodipine

DOSAGE FORM:

Tablet

THERAPEUTIC CLASS:

Anti hypertensive

Anti anginal

DESCRIPTION:

Amtas 5, 10: White to off white, round, flat, uncoated tablets with bevelled edges having breakline on one side and plain on other side.

PHARMACOLOGY

PHARMACODYNAMICS:

Mechanism of Action

Amlodipine is a dihydropyridine calcium antagonist (calcium ion antagonist or slow channel blocker) that inhibits the transmembrane influx of calcium ions into vascular smooth muscle and cardiac muscle. Experimental data suggests that Amlodipine binds to both dihydropyridine and nondihydropyridine binding sites. The contractile processes of cardiac muscle and vascular smooth muscle are dependent upon the movement of extracellular calcium ions into these cells through specific ion channels. Amlodipine inhibits calcium ion influx across cell membranes selectively, with a greater effect on vascular smooth muscle cells than on cardiac muscle cells. Negative inotropic effects can be detected in vitro but such effects have not been seen in intact animals at therapeutic doses. Serum calcium concentration is not affected by Amlodipine. Within the physiological pH range, Amlodipine is an ionized compound ($pK_a = 8.6$), and its kinetic interaction with the calcium channel receptor is characterised by a gradual rate of association and dissociation with the receptor binding site, resulting in a gradual onset of effect.

Amlodipine is a peripheral arterial vasodilator that acts directly on vascular smooth muscle to cause a reduction in peripheral vascular resistance and reduction in blood pressure.

The precise mechanisms by which Amlodipine relieves angina have not been fully delineated, but are thought to include the following :

Exertional Angina : In patients with exertional angina, Amlodipine reduces the total peripheral resistance (after load) against which the heart works and reduces the rate pressure product, and thus myocardial oxygen demand, at any given level of exercise.

Vasospastic Angina : Amlodipine has been demonstrated to block constriction and restore blood flow in coronary arteries in response to calcium, potassium epinephrine, serotonin and thromboxane A_2 analog in experimental animal models and in human coronary spasm is responsible for the effectiveness of Amlodipine in vasospastic (Prinzmetal's or variant) angina.

PHARMACOKINETICS:

After oral administration of therapeutic doses of Amlodipine, absorption produces peak plasma concentrations between 6 and 12 hours. Absolute bioavailability has been estimated to be between 64 and 90%. The bioavailability of amlodipine is not altered by the presence of food. Amlodipine is extensively (about 90%) converted to inactive metabolites via hepatic metabolism with 10% of the parent compound and 60% of the metabolites excreted in the urine. Ex vivo studies have shown that approximately 93% of the circulating drug is bound to plasma proteins in hypertensive patients. Elimination from the plasma is biphasic with a terminal elimination half-life of about 30-50 hours. Steady state plasma levels of Amlodipine are reached after 7 to 8 days of consecutive daily dosing.

INDICATIONS:

Amlodipine is indicated for the first line treatment of hypertension and can be used as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of amlodipine, which has been used in combination with a thiazide diuretic, alpha blockers, beta adrenoreceptor blocking agent, or an angiotensin-converting enzyme inhibitor.

Amlodipine is indicated for the first line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature.

Amlodipine may be used where the clinical presentation suggests a possible vasospastic/vasoconstrictive component but where vasospasm/vasoconstriction has not been confirmed.

Amlodipine may be used alone, as monotherapy, or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of beta blockers.

CONTRAINDICATIONS:

Amlodipine is contraindicated in patients with a known sensitivity to dihydropyridines, amlodipine, or any of the inert ingredients.

SIDE EFFECTS / ADVERSE REACTIONS:

Amlodipine is well tolerated.

The most commonly observed side effects were

Autonomic nervous system: flushing

Body as a whole: fatigue

Cardiovascular, general: edema

Central & peripheral nervous system: dizziness, headache

Gastrointestinal: abdominal pain, nausea

Heart rate/rhythm: palpitations

Psychiatric: somnolence

Less commonly observed side effects in marketing experience include

Autonomic nervous: dry mouth, increased sweating

Body as a whole: asthenia, back pain, malaise, pain, weight increase/decrease

Cardiovascular, general: hypotension, syncope

Central & peripheral nervous: hypertonia, hypoesthesia/paresthesia, peripheral neuropathy, tremor

Endocrine: gynecomastia

Gastrointestinal: altered bowel habits, dyspepsia (including gastritis), gingival hyperplasia, pancreatitis, vomiting

Metabolic/nutritional: hyperglycemia

Musculoskeletal: arthralgia, muscle cramps, myalgia

Platelet/bleeding/clotting: purpura, thrombocytopenia

Psychiatric: impotence, insomnia, mood changes

Respiratory: coughing, dyspnea, rhinitis

Skin/appendages: alopecia, skin discoloration, urticaria

Special senses: taste perversion, tinnitus

Urinary: increased urinary frequency, micturition disorders, nocturia

Vascular (extracardiac): vasculitis

Vision: visual disturbances

White blood cells/RES: leucopenia

Rarely, allergic reaction including pruritus, rash, angioedema, and erythema multiforme.

Hepatitis, jaundice and hepatic enzyme elevations have also been reported very infrequently (mostly consistent with cholestasis). Some cases severe enough to require hospitalization have been reported in association with use of amlodipine. In many instances, causal association is uncertain.

As with other calcium channel blockers, the following adverse events have been rarely reported and cannot be distinguished from the natural history of the underlying disease:

myocardial infarction, arrhythmia (including ventricular tachycardia and atrial fibrillation) and chest pain.

PRECAUTIONS / WARNINGS:

Use in patients with heart failure

Amlodipine was associated with increased reports of pulmonary edema despite no significant difference in the incidence of worsening heart failure as compared to placebo (see section pharmacodynamic properties)

Use in patients with impaired hepatic function

As with all calcium antagonist, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.

PREGNANCY AND LACTATION:

Safety of amlodipine in human pregnancy or lactation has not been established. Amlodipine does not demonstrate toxicity in animal reproductive studies other than to delay parturition and prolong labor in rats at a dose level fifty times the maximum recommended dose in humans. Accordingly, use in pregnancy is only recommended when there is no safer alternative and when the disease itself carries greater risk to the mother and fetus

DRUG INTERACTIONS:

Amlodipine has been safely administered with thiazide diuretics, alpha blockers, beta blockers, angiotensin-converting enzyme inhibitors, long-acting nitrates, sublingual nitroglycerine, anti-inflammatory drugs, antibiotics and oral hypoglycemic drugs.

In vitro data from studies with human plasma indicate that amlodipine has no effect on protein binding of the drugs tested (digoxin, phenytoin, warfarin or indomethacin).

In the following studies listed below, there were no significant changes in the pharmacokinetics of either amlodipine or another drug within the study, when co-administered.

Special studies: Effect of other agents on amlodipine

Cimetidine: Co-administration of amlodipine with cimetidine did not alter the pharmacokinetics of amlodipine.

Grapefruit juice: Co-administration of 240 ml of grapefruit juice with a single oral dose of amlodipine 10 mg in 20 healthy volunteers had no significant effect on the pharmacokinetics of amlodipine.

Aluminium/Magnesium (antacid): Co-administration of an aluminium/magnesium antacid with a single dose of amlodipine had no significant effect on the pharmacokinetics of amlodipine.

Sildenafil: A single 100 mg dose of sildenafil in subjects with essential hypertension had no effect on the pharmacokinetics parameters of amlodipine. When amlodipine and sildenafil were used in combination, each agent independently exerted its own blood pressure lowering effects.

Special studies: Effect of amlodipine on other agents

Atorvastatin: Co-administration of multiple 10 mg doses of amlodipine with 80 mg of atorvastatin resulted in no significant change in the steady state pharmacokinetic parameters of atorvastatin.

Digoxin: Co-administration of amlodipine with digoxin did not change serum digoxin levels or digoxin renal clearance in normal volunteers.

Ethanol (alcohol): Single and multiple 10 mg doses of amlodipine had no significant effect on the pharmacokinetics of ethanol.

Warfarin: Co-administration of amlodipine with warfarin did not change the warfarin prothrombin response time.

Cyclosporin: Pharmacokinetics studies with cyclosporins have demonstrated that does not significantly alter the pharmacokinetics of cyclosporin.

Drug/Laboratory Test Interactions: Not known

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:

For both hypertension and angina the usual initial dose is 5 mg amlodipine once daily which may be increased to a maximum dose of 10 mg depending on the individual patient's response.

No dose adjustment of amlodipine is required upon concomitant administration of thiazide diuretics, beta-blockers, and angiotensin-converting enzyme inhibitors.

Use in the elderly

Normal dosage regimens are recommended. Amlodipine used at similar doses in elderly or younger patients, is equally well tolerated.

Use in children

Safety and effectiveness of amlodipine in children have not been established.

Use in patients with impaired hepatic function

See section warnings and precautions

Use in renal failure

Amlodipine may be used in such patients at normal doses. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine is not dialyzable.

Route of administration:

Oral

OVERDOSE:

Available data suggest that gross overdosage could result in excessive peripheral vasodilatation and possibly reflex tachycardia. Marked and probably prolonged systematic hypotension up to end including shock with fatal outcome have been reported.

Administration of activated charcoal to healthy to volunteers immediately after or up to two hours of amlodipine 10 mg ingestion has been shown to significantly decrease amlodipine absorption.

Gastric lavage may be worthwhile in some cases.

Clinically significant hypotension due to amlodipine overdosage calls for active cardiovascular support including frequent monitoring of cardiac and respiratory function, elevation of extremities, and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockers. Since amlodipine is highly protein-bound, dialysis is not likely to be benefit.

STORAGE:

Store below 30°C.

PACK SIZE:

Alu/Alu blister of 3 x 10 / box

SHELF LIFE:

Two years from the date of Manufacturing.

Manufactured by:

INTAS

INTAS PHARMACEUTICALS LTD.

Plot No. 457 & 458, Village Matoda, Bavla road,
And Plot No. 191/218 P, Village Chacharwadi,
Ta -Sanand, Dist. Ahmedabad - 382210, India

Product Registration Holder:

Jetpharma Sdn Bhd, Selangor - Malaysia.

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